



K. J. Somaiya College of Engineering, Mumbai-77
(A Constituent College of Somaiya Vidyavihar University)
Department of Computer Engineering

Batch: B3 Roll No.: 1611124

Experiment No. _____

Grade: AA / AB / BB / BC / CC / CD / DD

Signature of the Staff In-charge with date

Title: Implementation of sum of subset Algorithm

Objective: To learn the Backtracking strategy of problem solving for Sum of subset

CO to be achieved:

CO 2 Analyze and solve problems for divide and conquer strategy, greedy method, dynamic programming approach and backtracking and branch & bound policies.

Books/ Journals/ Websites referred:

1. Ellis horowitz, Sarataj Sahni, S.Rajsekaran," Fundamentals of computer algorithm", University Press
2. T.H.Cormen ,C.E.Leiserson,R.L.Rivest and C.Stein," Introduction to algortihmts",2nd Edition ,MIT press/McGraw Hill,2001

Pre Lab/ Prior Concepts:

Data structures, Concepts of algorithm analysis

Historical Profile:

Subset sum problem is to find subset of elements that are selected from a given set whose sum adds up to a given number K. We are considering the set contains non-negative values. It is assumed that the input set is ~~unique (no duplicates are presented)~~.

One way to find subsets that sum to K is to consider all possible subsets. A power set contains all those subsets generated from a given set. The size of such a power set is 2^N .

Input:

A vector $X=\{x_1, x_2, \dots, x_n\}$ for all n elements in the set where $X_i=0$ (element not added) or $x_i=1$ (element added in the solution tuple).

Output:

Summation of the chosen numbers must be equal to given number M and one number can be used only once.

BACKTRACKING CONDITION



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$$B_k(x_1, \dots, x_k) = \text{true} \text{ iff } \sum_{i=1}^k w_i x_i + \sum_{i=k+1}^n w_i \geq m$$
$$\text{and } \sum_{i=1}^k w_i x_i + w_{k+1} \leq m$$

New Concepts to be learned:

Application of algorithmic design strategy to any problem, Backtracking method of problem solving Vs other methods of problem solving problem sum of subset and its applications.

Algorithm:

Algorithm sumOfSub(s, k, r)

{//It is assumed $w[1] \leq m$ and $\sum_{i=1}^m w[i] \geq m$

//generate the left child. Note: $s + w(k) \leq M$ since B_{k-1} is true.

$X[k] = 1;$

if $(S + W[k] = m)$ then write($X[1:k]$); //Subset found. there is no recursive call here
as $W[j] > 0, 1 \leq j \leq n$.

else if $(S + W[k] + W[k+1] \leq m)$ then sumOfSub($S + W[k], k+1, r - W[k]$); //moving to
next sub-problem.

Similarly, assume the array is presorted and we found one subset. We can generate
next node excluding the present node only when inclusion of next
node satisfies the constraints.

if $((S + r - W[k] \geq m) \text{ and } (S + W[k+1] \leq m))$ then //generate right {
//child and those satisfying 2 bounding functions

$X[k] = 0;$

sumOfSub ($S, k+1, r - W[k]$);
}}

Implementation(Code):



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Output:

Example sum of subset Problem along with state space tree:

Analysis of Backtracking solution for sum of subset Problem:

Conclusion: